

Asignatura

Sistemas secuenciales programables

UNIDAD 1

Lógica digital (II)

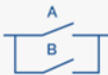
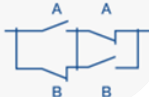


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Instituto Superior de FP

Funciones lógicas

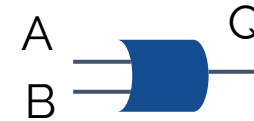
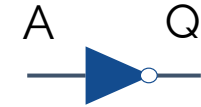


Ejemplos circuitos lógicos

	Tabla de la verdad	Símbolo IEC	Símbolo ANSI	Ecuación lógica	Circuito equivalente															
DIRECTA	<table><tr><td>A</td><td>Q</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr></table> 	A	Q	0	0	1	1			$Q = A$										
A	Q																			
0	0																			
1	1																			
NOT	<table><tr><td>A</td><td>Q</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	Q	0	1	1	0			$Q = \bar{A}$										
A	Q																			
0	1																			
1	0																			
AND (Y)	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> 	A	B	Q	0	0	0	0	1	0	1	0	0	1	1	1			$Q = A \cdot B$	
A	B	Q																		
0	0	0																		
0	1	0																		
1	0	0																		
1	1	1																		
OR (O)	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Q	0	0	0	0	1	1	1	0	1	1	1	1			$Q = A + B$	
A	B	Q																		
0	0	0																		
0	1	1																		
1	0	1																		
1	1	1																		
NAND	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	1	0	1	1	1	0	1	1	1	0			$Q = \overline{A \cdot B}$	
A	B	Q																		
0	0	1																		
0	1	1																		
1	0	1																		
1	1	0																		
NOR	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	1	0	1	0	1	0	0	1	1	0			$Q = \overline{A + B}$	
A	B	Q																		
0	0	1																		
0	1	0																		
1	0	0																		
1	1	0																		
XOR	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	0	0	1	1	1	0	1	1	1	0			$Q = A \oplus B$	
A	B	Q																		
0	0	0																		
0	1	1																		
1	0	1																		
1	1	0																		
NXOR	<table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Q	0	0	1	0	1	0	1	0	0	1	1	1			$Q = \overline{A \oplus B}$	
A	B	Q																		
0	0	1																		
0	1	0																		
1	0	0																		
1	1	1																		

A	Q
0	1
1	0

NOT



OR



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1



Ecuación a partir de tabla de la verdad

Minterms

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$F = \bar{A} * \bar{B} * \bar{C} + \bar{A} * \bar{B} * C + \bar{A} * B * C + A * \bar{B} * C + A * B * C$$

Maxterms

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$F = (A + \bar{B} + C) * (\bar{A} + B + C) * (\bar{A} + B + \bar{C}) * (\bar{A} + \bar{B} + C)$$

Tabla verdad a MK (minterms)



A	B	C	D	Q
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	1	0	0
1	1	0	1	1
1	1	1	1	0

Mapa de Karnaugh

AB/CD	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	1	0	0	1
10	1	0	0	1

2^n

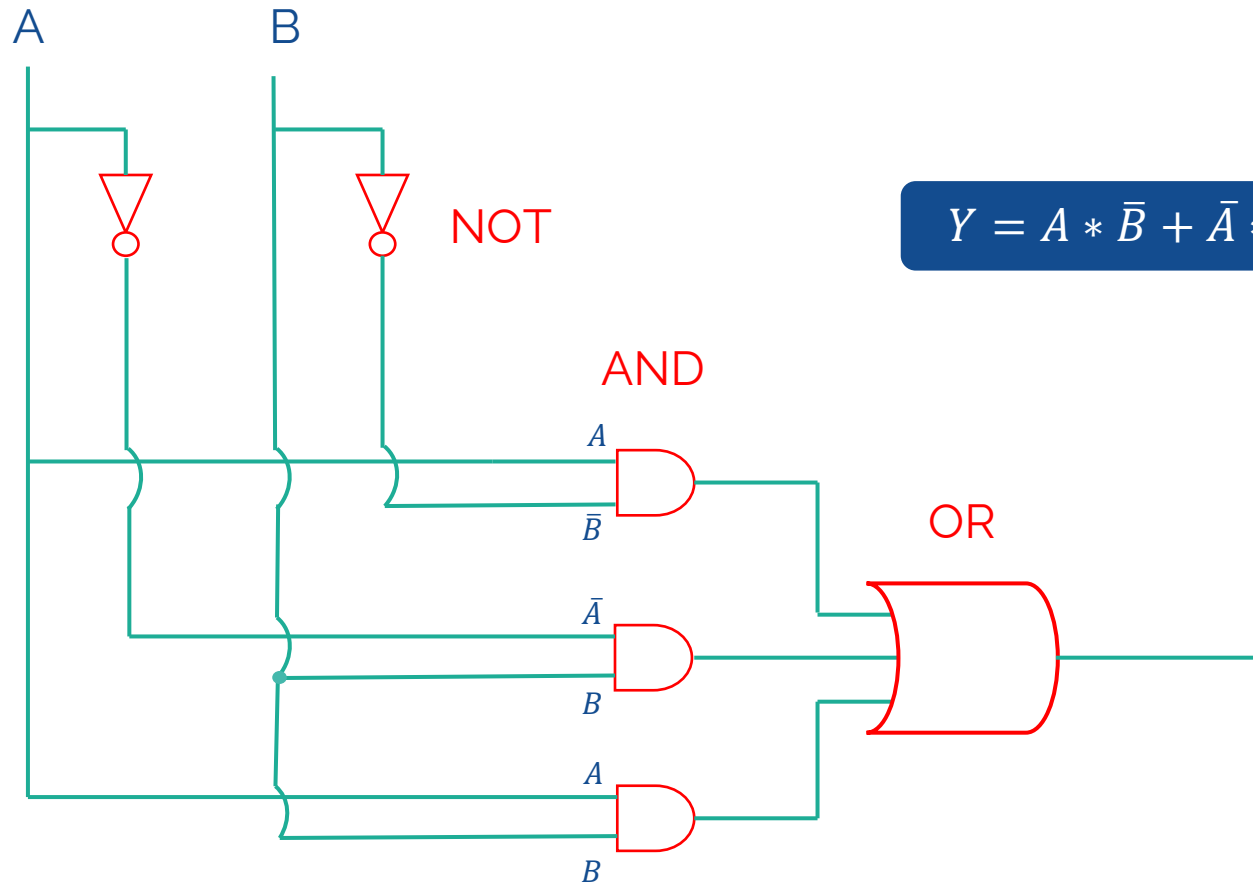
$\bar{A}\bar{B}\bar{C}\bar{D}$		$\bar{D}$
$\bar{A}\bar{B}\bar{C}D$		
$\bar{A}\bar{B}C\bar{D}$		
$\bar{A}\bar{B}CD$		
$\bar{A}B\bar{C}\bar{D}$		$\bar{A}B$
$\bar{A}B\bar{C}D$		
$\bar{A}BC\bar{D}$		
$\bar{A}BCD$		

$$Q = \bar{A} * B + \bar{D}$$

Leyes de Morgan

$$Q = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD$$

Ecuación a partir de un circuito



$$Y = A * \bar{B} + \bar{A} * B + A * B$$

Circuitos secuenciales y combinacionales



Sistemas combinacionales

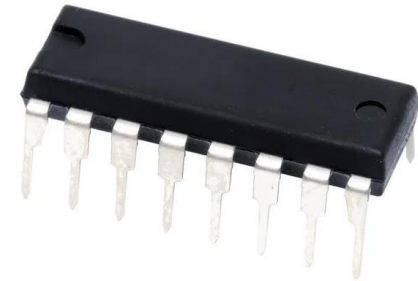
Dependen
solo del
estado de su
salida

- Multiplexores
- Demultiplexores
- Codificadores
- Decodificadores

Sistemas secuenciales

Dependen de
las entradas y
el estado
anterior de
ellas mismas

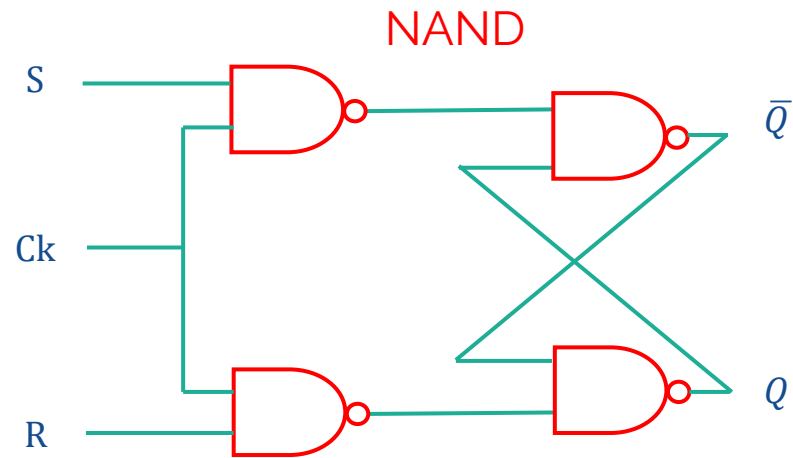
BIESTABLES



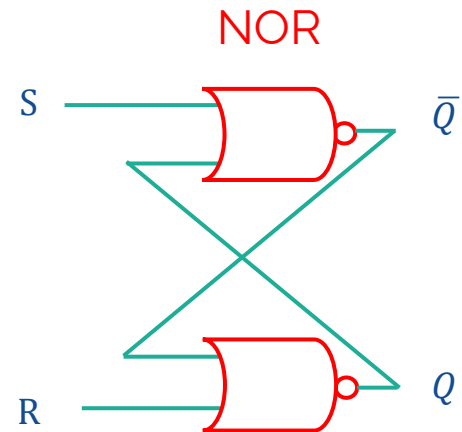
El biestable (flip-flop)

Tiene dos estados estables y almacena la información pasada (estado)

Biestable RS síncrono



Biestable RS asíncrono





El biestable RS asíncrono

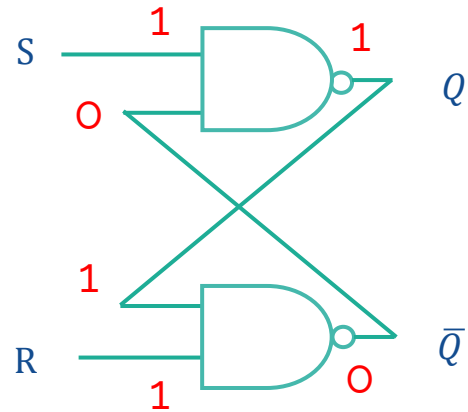
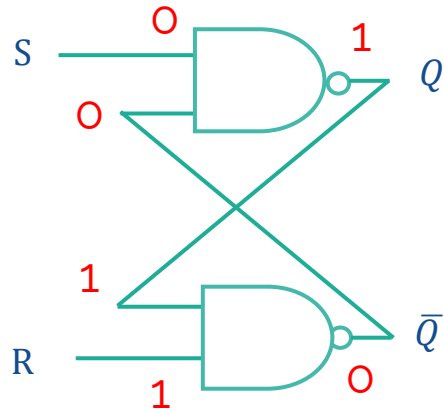
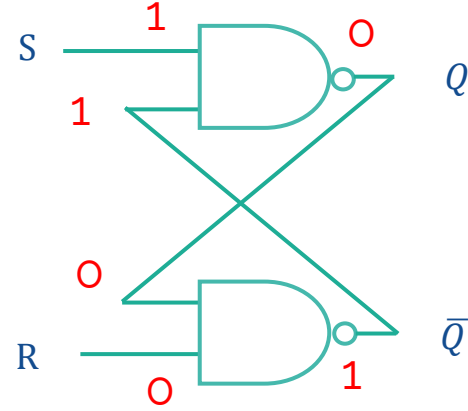
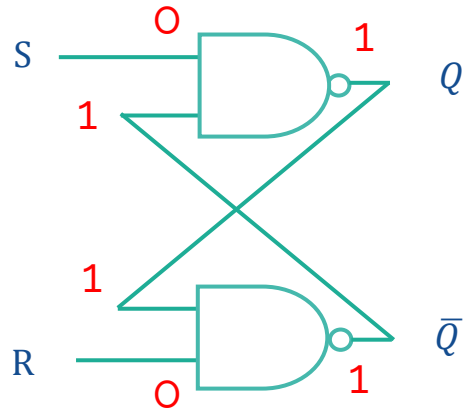
NAND

A	B	Q
0	0	1
1	0	1
0	1	1
1	1	0



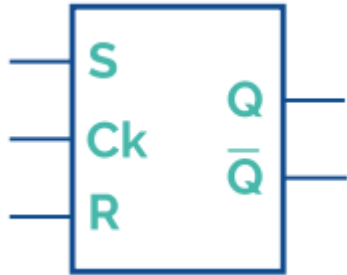
S	R	Q	$\bar{Q}$
0	0	x	x
1	0	0	1
0	1	1	0
1	1	1	0

EP
Reset
Set
Man

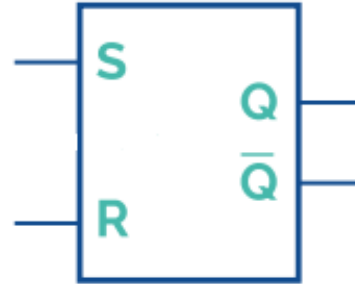


La simbología de los biestables

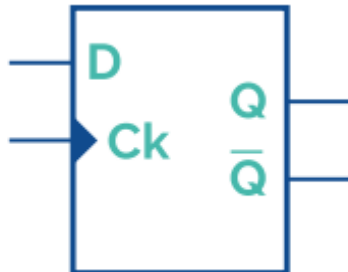
Biastable RS síncrono



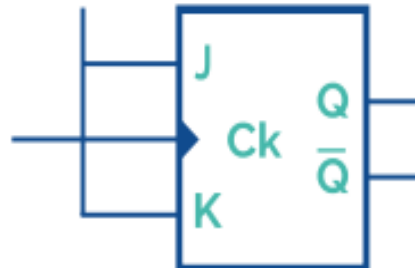
Biastable RS asíncrono



Biastable D



Biastable síncrono T



The background is a solid blue color with a subtle, large-scale grid pattern. Overlaid on this are numerous small, light blue arrows pointing in various directions, creating a sense of movement and flow. The overall aesthetic is modern and technological.

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